

SECTION 15048

MECHANICAL PAINTING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.
- B. Referenced Specifications:
 - 1. Section 09800 - Special Coatings

1.2 SUMMARY

- A. This Section includes surface preparation, painting, and finishing of building exposed exterior and interior mechanical scope items.
 - 1. Surface preparation, priming, and finish coats specified in this section are in addition to shop priming and surface treatment specified under other sections.
- B. Paint exposed surfaces whether or not colors are designated in "schedules," except where a surface or material is specifically indicated not to be painted or is to remain natural. Where an item or surface is not specifically mentioned, paint the same as similar adjacent materials or surfaces. If color or finish is not designated, the Architect/Engineer will select from standard colors or finishes available.
 - 1. Painting scope includes field painting of un-insulated pipes in conditioned spaces supplying Cooling Coils and Air Handling Units, hangers, exposed steel and iron work, and primed metal surfaces of mechanical equipment.
- C. Painting is not required on pre-finished items, finished metal surfaces, concealed surfaces, operating parts, and labels.
 - 1. Pre-finished items not to be painted include the following factory-finished components:
 - a. Finished mechanical equipment
 - b. Air diffusers, grills and registers
 - 2. Operating parts not to be painted include moving parts of operating equipment such as the following:
 - a. Valve and damper operators.
 - b. Linkages.
 - c. Sensing devices.
 - d. Motor and fan shafts.
 - 3. Labels: Do not paint over Underwriter's Laboratories, Factory Mutual or other code-required labels or equipment name, identification, performance rating, or nomenclature plates.

1.3 SUBMITTALS

- A. Product Data: Manufacturer's technical information, label analysis, and application instructions for each material proposed for use.
 - 1. List each material and cross-reference the specific coating and finish system and application. Identify each material by the manufacturer's catalog number and general classification.
- B. Samples for verification purposes: Provide samples of each color and material to be applied, with texture to simulate actual conditions, on representative samples of the actual substrate. Define each separate coat, including block fillers and primers. Use representative colors when preparing samples for review. Resubmit until required sheen, color, and texture are achieved.
 - 1. Provide a list of material and application for each coat of each sample. Label each sample as to location and application.
 - 2. Submit samples on the following substrates for the Architect/Engineer's review of color and texture only:
 - a. Ferrous Metal: Provide two 4-inch-square samples of flat metal and two 8-inch-long samples of solid metal for each color and finish.
- C. Substitutions:
 - 1. Requests for Substitutions: Submit three copies, fully identified for product being replaced by substitution, including related specification section and drawing number(s), and fully documented to show compliance with requirements for substitutions. Include product data/drawings; description of methods; colors, textures and patterns; samples where applicable.

1.4 QUALITY ASSURANCE

- A. Single-Source Responsibility: Provide primers and undercoat paint produced by the same manufacturer as the finish coats.
- B. Coordination of Work: Review other sections in which primers are provided to ensure compatibility of the total systems for various substrates. On request, furnish information on characteristics of finish materials to ensure use of compatible primers.
 - 1. Notify the Architect/Engineer of problems anticipated using the materials specified.
- C. Material Quality: Provide the manufacturer's best quality trade sale paint material of the various coating types specified. Paint material containers not displaying manufacturer's product identification will not be acceptable.
 - 1. Proprietary names used to designate colors or materials are not intended to imply that products named are required or to exclude equal products of other manufacturers.

1.5 JOB CONDITIONS

- A. Apply water-based paints only when the temperature of surfaces to be painted and surrounding air temperatures are between 50 deg F (10 deg C) and 90 deg F (32 deg C).
- B. Apply solvent-thinned paints only when the temperature of surfaces to be painted and surrounding air temperatures are between 45 deg F (7 deg C) and 95 deg F (35 deg C).

C. Do not apply paint in snow, rain, fog, or mist, when the relative humidity exceeds 85 percent, at temperatures less than 5 deg F (3 deg C) above the dew point, or to damp or wet surfaces.

1. Painting may continue during inclement weather if surfaces and areas to be painted are enclosed and heated within temperature limits specified by the manufacturer during application and drying periods.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Available Manufacturers: Subject to compliance with requirements, manufacturers offering products that may be incorporated in the work include but are not limited to the following:

1. Tnemec Company, Inc.
2. Approved equivalent

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates and conditions under which painting will be performed for compliance with requirements for application of paint. Do not begin paint application until unsatisfactory conditions have been corrected.

1. Start of painting will be construed as the Applicator's acceptance of surfaces and conditions within a particular area.

3.2 PREPARATION

A. General Procedures: Remove hardware and hardware accessories, plates, machined surfaces, and similar items in place that are not to be painted, or provide surface-applied protection prior to surface preparation and painting. Remove these items if necessary for complete painting of the items and adjacent surfaces. Following completion of painting operations in each space or area, have items reinstalled by workers skilled in the trades involved.

1. Clean surfaces before applying paint or surface treatments. Remove oil and grease prior to cleaning. Schedule cleaning and painting so that dust and other contaminants from the cleaning process will not fall on wet, newly painted surfaces.

B. Surface Preparation: Clean and prepare surfaces to be painted in accordance with the manufacturer's instructions for each particular substrate condition and as specified.

1. Provide barrier coats over incompatible primers or remove and re-prime. Notify Architect/Engineer in writing of problems anticipated with using the specified finish-coat material with substrates primed by others.
2. Ferrous Metals: Clean non-galvanized ferrous-metal surfaces that have not been shop coated; remove oil, grease, dirt, loose mill scale, and other foreign substances. Use solvent or mechanical cleaning methods that comply with recommendations of the Steel Structures Painting Council.
 - a. Treat bare and sandblasted or pickled clean metal with a metal treatment wash coat before priming.

- b. Touch up bare areas and shop-applied prime coats that have been damaged. Wire-brush, clean with solvents recommended by the paint manufacturer, and touch up with the same primer as the shop coat.
- 3. Galvanized Surfaces: Clean galvanized surfaces with non-petroleum-based solvents so that the surface is free of oil and surface contaminants. Remove pretreatment from galvanized sheet metal fabricated from coil stock by mechanical methods.
- C. Materials Preparation: Carefully mix and prepare paint materials in accordance with manufacturer's directions.
 - 1. Maintain containers used in mixing and application of paint in a clean condition, free of foreign materials and residue.
 - 2. Stir material before application to produce a mixture of uniform density; stir as required during application. Do not stir surface film into material. Remove film and, if necessary, strain material before using.
 - 3. Use only thinners approved by the paint manufacturer, and only within recommended limits.
- D. Tinting: Tint each undercoat a lighter shade to facilitate identification of each coat where multiple coats of the same material are applied. Tint undercoats to match the color of the finish coat, but provide sufficient differences in shade of undercoats to distinguish each separate coat.

3.3 APPLICATION

- A. Apply paint in accordance with manufacturer's directions. Use applicators and techniques best suited for substrate and type of material being applied.
- B. Do not paint over dirt, rust, scale, grease, moisture, scuffed surfaces, or conditions detrimental to formation of a durable paint film.
- C. Paint colors, surface treatments, and finishes are indicated in Paragraph 3.6 "Paint Schedules."
- D. Provide finish coats that are compatible with primers used.
- E. The number of coats and film thickness required is the same regardless of the application method. Do not apply succeeding coats until the previous coat has cured as recommended by the manufacturer. Sand between applications, where sanding is required, to produce an even smooth surface in accordance with the manufacturer's directions.
- F. Apply additional coats when undercoats, stains, or other conditions show through final coat of paint until paint film is of uniform finish, color, and appearance. Give special attention to ensure that surfaces, including edges, corners, crevices, welds, and exposed fasteners, receive a dry film thickness equivalent to that of flat surfaces.
- G. The term "exposed surfaces" includes areas visible when permanent or built-in fixtures, convector covers, grilles, and similar components are in place. Extend coatings in these areas as required to maintain the system integrity and provide desired protection.
- H. Paint surfaces behind movable equipment and furniture the same as similar exposed surfaces. Paint surfaces behind permanently fixed equipment or furniture with prime coat only before final installation of equipment.
- I. Paint back sides of access panels and removable or hinged covers to match exposed surfaces.
- J. Omit primer on metal surfaces that have been shop-primed and touch up painted.

K. Scheduling Painting: Apply first coat to surfaces that have been cleaned, pretreated, or otherwise prepared for painting as soon as practicable after preparation and before subsequent surface deterioration.

1. Allow sufficient time between successive coats to permit proper drying. Do not recoat until paint has dried to where it feels firm, and does not deform or feel sticky under moderate thumb pressure and where application of another coat of paint does not cause lifting or loss of adhesion of the undercoat.

M. Minimum Coating Thickness: Apply materials at not less than the manufacturer's recommended spreading rate. Provide a total dry film thickness of the entire system as recommended by the manufacturer.

N. Mechanical items to be painted include but are not limited to:

1. Uninsulated chilled water piping in conditioned spaces serving Sensible Cooling Coils and
2. Ductwork.
3. Insulation.
4. Supports, including contractor furnished supplemental steel.

3.4 CLEANING

A. Cleanup: At the end of each work day, remove empty cans, rags, rubbish, and other discarded paint materials from the site.

B. Upon completion of painting, clean glass and paint-spattered surfaces. Remove spattered paint by washing and scraping, using care not to scratch or damage adjacent finished surfaces.

3.5 PROTECTION

A. Protect work of other trades, whether to be painted or not, against damage by painting. Correct damage by cleaning, repairing or replacing, and repainting, as acceptable to Architect/Engineer. Provide "wet paint" signs to protect newly painted finishes. Remove temporary protective wrappings provided by others for protection of their work after completion of painting operations.

B. At completion of construction activities of other trades, touch up and restore damaged or defaced painted surfaces.

3.6 PAINT SCHEDULE

A. For pipe support systems refer to Specification Section 09800 - Special Coatings.

B. General: Provide the following paint systems for the various substrates indicated.

C. Ferrous Metal: Primer is not required on shop-primed items.

1. Epoxy-Polyamide High-Build Acrylic Polyurethane Enamel Semi-Gloss Finish:

- a. Surface Preparation: SSPC-SP6 Commercial Blast Cleaning.
- b. Shop Primed Surfaces: Shop prime with Tnemec Series 66-AA90 Hi-Build Epoxoline, 4.0 to 6.0 mils DFT.
 - 1) Touch-Up: Power tool clean spots and touch up with Tnemec Series 66-AA90 Hi-Build Epoxoline.
 - 2) Second Coat: Tnemec Series 73 - Color Endura-Shield III, 3.0 to 5.0 mils DFT.
 - 3) Third Coat: Tnemec Series 76 - Gloss Clear, 1.5 mils DFT.

c. Field Primed Surfaces:

- 1) Surface Preparation: SSPC-SP6 Commercial Blast Cleaning.
- 2) Primer: Tnemec Series 66-AA90 Hi-Build Epoxoline, 4.0 to 6.0 mils DFT.
- 3) Second Coat: Tnemec Series 73 - Color Endura-Shield III, 3.0 to 5.0 mils DFT.
- 4) Third Coat: Tnemec Series 76 - Gloss Clear, 1.5 mils DFT.

D. Zinc-Coated Metal (Galvanized Steel):

1. Semi-gloss Smooth Finish: Primer and finish coat with film thickness of 4.0 mils DFT.

- a. Surface Preparation: SSPC SP1, Solvent Cleaning.
- b. Primer: Tnemec Series 27 F.C. Typoxy, 2.0 to 3.5 mils DFT.
- c. Finish Coat: Tnemec Series 73 Endura-Shield, 2.0 mils DFT.

E. Metal Piping:

1. Epoxy-Polyamide Acrylic Polyurethane Coating

- a. Satin Finish
- b. Exposure: Dry interior conditions.
- c. Surface Prep: SSPC-SP6 Commercial Blast Cleaning. For areas that will be constantly wet or damp, or for sweating pipes, change surface prep to SSPC-SP10 Near White Metal Blast Cleaning.
- d. Coating System:
 - 1st Coat: Tnemec Series 161 Tneme-Fascure 3.0 - 5.0 mils
 - 2nd Coat: Tnemec Series 73-Color Endura-Shield III 3.0 - 5.0 mils

END OF SECTION